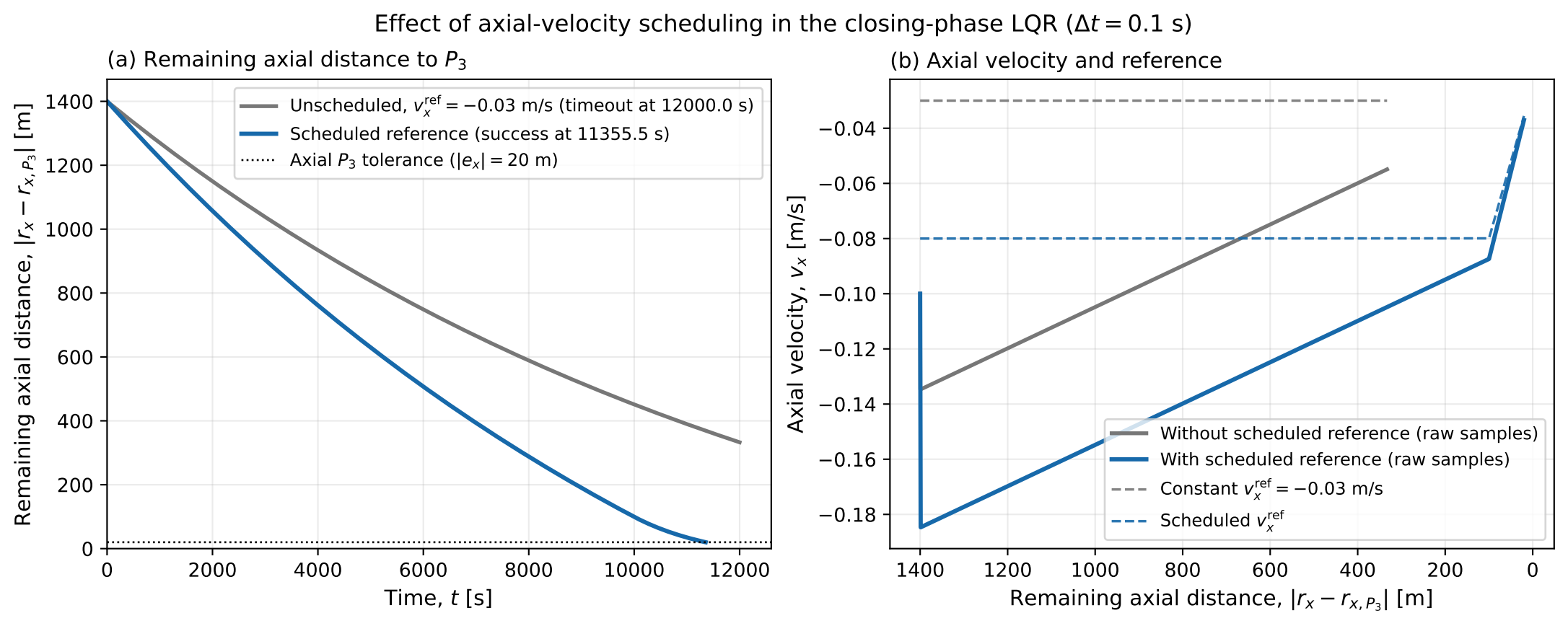
Effect of axial-velocity scheduling in the closing-phase LQR ($\Delta t = 0.1$ s)
(a) Remaining axial distance to P_3
Remaining axial distance, $|r_x - r_{x,P_3}|$ [m]
Time, t [s]
Unscheduled, $v_x^{\mathrm{ref}} = -0.03$ m/s (timeout at 12000.0 s)
Scheduled reference (success at 11355.5 s)
Axial P_3 tolerance ($|e_x| = 20$ m)
(b) Axial velocity and reference
Axial velocity, v_x [m/s]
Remaining axial distance, $|r_x - r_{x,P_3}|$ [m]
Without scheduled reference (raw samples)
With scheduled reference (raw samples)
Constant $v_x^{\mathrm{ref}} = -0.03$ m/s
Scheduled v_x^{ref}